Modeling the Dynamics of Learning:
Comparing Three Architectures of Artificial Cognition

Why this work?

- The rise of AI in Education has brought prediction, recommendation, and personalization into every learning platform.
- Yet, most systems still rely on surface metrics (clicks, activity, completio) – not true learning dynamics.

The challenge:

How can we design AI models that understand the cogniitive process of learning, not just its traces?

Scientific question:

Can different architectures of intelligence – statistical, generative, and cognitive – predict and explain how humans learn over time?

Modeling the Dynamics of Learning

Observation

- Learning is a sequential and contextual process, not a series of independent events
- Plattforms capture clicks, not cognitive trajectory

Objective:

Compare three views of memory and reasoning to predict the educational journey

MEMORY → REASONING → HIERARCHY

Three Models, Three Ways of Reasoning

Paradigm	Model	Key Mechanism	Strengths	Limitations
Dynamic Memory	DKVMN (baseline)	Updates a key-value skill map after each interactiction	Frugal, stable, explainable	No symbolic reasoning
Explicit Reasoning	LLM + CoT (e.g.. Qwen 2.5 / Mistral 7B)	Generates intermediate reasoning steps ("chain of thought") before prediction	Interpretable, flexible, semantic transfer	Cost, latency, difficult to calibrate
Hierarchical Memory	HRM (prototype)	Three memories: short-, medium-and long-term + retrieval	Temporal structure, controlled forgetting, multi-source reasoning	Complexity, memory tuning

Comparative Objective: Which architecture best reproduces the dynamics of learning?

Experimental Evaluation

Data & Protocol

- Benchmarks: Riiid!, EdNet, ASSISTments
- Temporal split: train < val < test
- Inputs: question → answer sequences
- Targets:
  - Predict success on the next item
  - Anticipate learner disengagemennt
  - Estimate time-to-mastery

Metrics

Type	Metric	Interpretatioton
Accuracy	AUC / NDCG5/ Brier	Fidelity of predictions
Calibration	ECE / Brier decomposition	Model confidence
Sequence	c-index / IBS	Mastery of temprability
Explainability	Cof vs: memory weights. vs. SHAP	Human interpretability

Expected Comparison

- DKVMN: reliable baseline, flat memory
- LLM (CoT): strong reasoning, but higher cost
- HRM: balance: between cognition, frugality, and traceability

Deliverables

- Code: three trained & evaluated models (PyTorch / FastAPI)
- Dashboard: comparative visulization of performance and metrics
- Report: quantitative results + cognitive analysis + Model Cards (AI)